PAYTM SAFECLEAR

An Autonomous Incident Resolution Teammate for Ambiguous Point-of-Sale UPI Transactions

Track 3: Autonomous AI Teammates

By Team Ok Paytm

WHAT'S THE PROBLEM?

The Counter Friction:
When a UPI payment attempt results in an in-flight, pending, or timeout status, existing systems leave the counter in an operational vacuum:

- The Soundbox remains silent (it is architected to announce verified successes only).
- The customer fears an unconfirmed debit; the merchant cannot release goods without verification.

The Fallout:

- Accidental Duplicate Payments: Confused customers frequently scan and pay again, often creating a double debit and compounding dispute volume.
- Queue Delays & Manual Overhead: Merchants must interrupt their workflow to manually open their business app, locate the transaction, and refresh status.
- Delayed Recourse: While NPCI's UDIR framework and RBI guidelines resolve failed transactions over subsequent business days, there is no autonomous coordination at the physical point of sale during the incident itself.

PROPOSED SOLUTION

1. **Immediate Multi-Interface Communication:** When an ambiguous payment outcome occurs, SafeClear immediately issues clear audio and visual alerts on supported interfaces ("Payment verification in progress — please do not pay again yet"), defusing tension and preventing panic retries.
2. **Authoritative Status Orchestration:** Instead of requiring manual app polling, the agent coordinates throttled, authorized status inquiries against the acquiring switch and available status APIs.
3. **Duplicate Payment Guard:** Displays persistent visual holding states on the Paytm customer and merchant screens to discourage immediate re-scans.
4. **Automated Escalation & Docketing:** If status remains unresolved after verified retries, the agent pre-packages an authenticated dispute docket (capturing client reference, timestamps, and terminal telemetry) ready for authorized UDIR submission, or applies contractually defined, pre-funded merchant protection rules where available.

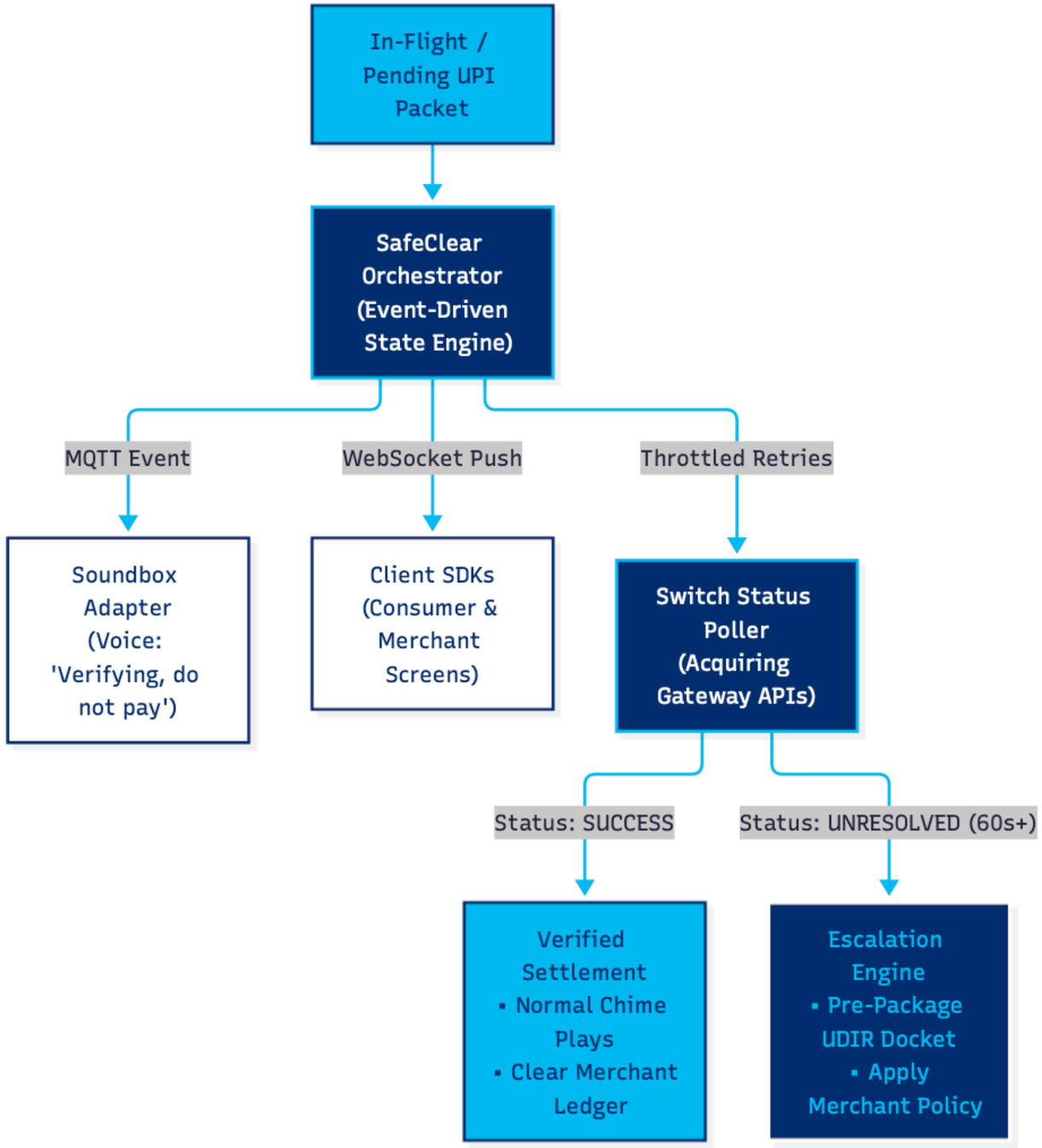
TECH STACK

• State Core (FastAPI + Redis)
Event-driven state machine managing active incidents with sub-100ms response time.

• Dual-Channel Counter Sync
Soundbox speaks warning (MQTT) + app screens lock (WebSockets) to stop duplicate payments.

• Authoritative Switch Poller
Automates background gateway status checks without manual merchant refreshing.

• Automated Dispute Docketing
Auto-compiles verified UTR/telemetry data for instant NPCI UDIR escalation after 60s.



USP

1. Orchestration Over Speculation: SafeClear does not gamble on artificial confidence scores or fake SMS receipts; it operates strictly on authoritative verification and defined policy rules.

2. Incident Lifecycle Ownership: Bridges the gap between the moment a payment hangs and when back-office dispute systems take over.

3. Active Duplicate Prevention: The first counter system designed specifically to prevent accidental double-spend attempts during network latency.

4. Zero Manual Investigation for Merchants: Turns an active queue interruption into an automated, background-monitored task.

IMPACT & BENEFITS

- **Reduced Resolution Time:** Eliminates manual navigation and status refreshing on the merchant app during peak rush hours.

- **Lowered Dispute Volume:** Preventing accidental duplicate payments directly reduces the downstream reconciliation workload for banks and customer care.

- **Preserved Trust:** Clear, automated communication eliminates awkward customer-merchant standoffs during bank server latency.

- **Audit-Ready Escalations:** Pre-packaged dispute dockets ensure that any escalation to UDIR contains validated transaction references and telemetry, cutting human dispute investigation time.

BUSINESS MODEL

Core Value for Paytm:

- Direct reduction in tier-1 customer support tickets related to in-flight payment inquiries and duplicate debits.
- High-value differentiator for the Paytm for Business ecosystem, reinforcing merchant reliance on Paytm hardware and apps.

PHASE 1

Software only: Real-time duplicate alerts on the Paytm Consumer App + automated status monitoring on the Paytm for Business App.

PHASE 2

Hardware Integration: Firmware adapter rollout to the Soundbox fleet for verbal incident de-escalation.

PHASE 3

Enterprise Merchant Protection: Optional integration with commercially backed merchant-protection programs for high-volume retail chains.